

Lab 05 — High-Performance Storage & Database Selection

Work with a relational database, an in-memory cache and object storage side by side, then choose the right store for each workload.

Lab environment: Docker Desktop (free local simulation) — runs locally, no AWS account needed

Estimated time: 35–40 minutes

SAA-C03 domain: Design High-Performing Architectures

Overview

Local component	AWS equivalent
lab5-db (MariaDB)	Amazon RDS / Aurora
lab5-redis (Redis)	Amazon ElastiCache
lab5-minio (MinIO)	Amazon S3

No ports are published for this lab — every exercise uses `docker exec`.

Step 1 — Get the lab source and start the lab

The repo lives in `~/Downloads/aws-saa`. The paths are absolute, so these commands work from any folder. `git clone` is needed only once: if the repo is already there it prints `already exists`, and the next line still takes you into this lab.

```
cd ~/Downloads
git clone https://github.com/mohanpothula/aws-saa.git
cd ~/Downloads/aws-saa/labs/lab-05-storage-database-selection
docker compose up -d
docker compose ps
```

MariaDB needs 10–20 seconds to initialise on first start. If Step 2 fails with a connection error, wait a moment and run it again.

Step 2 — SQL workload

Task: Create and query product data.

```
docker exec lab5-db mariadb -uroot -pPass123! labdb -e "CREATE TABLE products(id INT,name VARCHAR(30)); INSERT INTO products VALUES(1,'Laptop'); SELECT * FROM products;"
```

Expected result: `Laptop` is returned.

Step 3 — Cache workload

Task: Set and get a cached value.

```
docker exec lab5-redis redis-cli SET product:1 Laptop
docker exec lab5-redis redis-cli GET product:1
```

Expected result: `Laptop` is returned.

Step 4 — Object storage

Task: Create a bucket folder and upload a file.

```
docker exec lab5-minio sh -c 'mkdir -p /data/lab5 && echo report >/data/lab5/report.txt && ls -l /data/lab5'
```

Expected result: `report.txt` is listed.

Step 5 — Selection challenge

Task: Choose storage for a transaction, a session cache and an image archive.

```
echo 'Transaction=? Session=? Archive=?'
```

Expected result: Transaction → MariaDB (RDS) · Session → Redis (ElastiCache) · Archive → MinIO (S3).

Step 6 — Validate services

Task: Check all three services.

```
docker compose ps
```

Expected result: All three are running.

Verification

Check	Expected
Step 2 — SQL workload	<code>Laptop</code> is returned
Step 3 — Cache workload	<code>Laptop</code> is returned
Step 4 — Object storage	<code>report.txt</code> is listed
Step 5 — Selection challenge	Transaction → MariaDB (RDS) · Session → Redis (ElastiCache) · Archive → MinIO (S3)
Step 6 — Validate services	All three are running

Completion check

- ☐ Every validation result demonstrated to the trainer
- ☐ You can say which AWS service each local component represents
- ☐ You can justify each store in the selection challenge

Cleanup

```
docker compose down -v
```

`-v` also removes the lab's volumes, so the next run starts from a clean state.

Reference

- Lab source code:
<https://github.com/mohanpothula/aws-saa/tree/main/labs/lab-05-storage-database-selection>
- Original student guide: `courseware/AWS_SAA_Lab5_Student_Guide.docx`
- [Amazon RDS User Guide](#)
- [Amazon ElastiCache](#)
- [Amazon S3 User Guide](#)
- [Amazon S3 storage classes](#)
- [MariaDB official Docker image](#)
- [MinIO container documentation](#)